FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2024
FULL TEST – IX
PAPER –1
TEST DATE: 12-05-2024

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

1. B, D

Sol. Error $= \frac{1}{2}\text{ mm} + \frac{12.5}{100} \times 1\text{ mm} = 0.625\text{ mm}$

Measured value = 15 mm + $18 \times \frac{1}{100} + 0.625 = 15.805$ mm

2. A, C

Sol. Time between two consecutive collision = $\frac{10}{1000}$ sec

So, $\frac{2\ell}{v_{rms}} = \frac{10}{1000}$

$v_{rms} = 4000$ m/s

$v_{rms} = \sqrt{\frac{3RT}{M}} = 4000$

So, $T = \frac{16 \times 10^6 \times 4 \times 10^{-3}}{3 \times \left(\frac{25}{3}\right)} = 2560$ K

Mass of helium gas $\Rightarrow$ PV = nRT

$PV = \frac{m}{M}RT$

$m = \frac{PVM}{RT} = \frac{160 \times 8000 \times (4 \times 10^{-3})}{\left(\frac{25}{3}\right) \times 2560} = 240$ gm

3. A, C

Sol. Using conservation of momentum of the system

$P = P_2 \cos\theta$ …(i)

$P_1 = P_2 \sin\theta$ ….(ii)

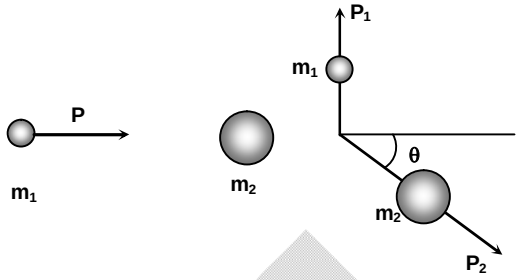
Solving (i) and (ii), we get

$P^2 + P_1^2 = P_2^2$

$2m_1k + 2m_1k_1 = 2m_2k_2$

$\therefore\ 4k_2 - k_1 = k$

$\Rightarrow\ 4k_2 - k_1 = 60$ …(iii)



Now $E_n = -13.6\left(\frac{Z^2}{n^2}\right)eV$

$\therefore E_1 = -54.4$ eV

$E_2 = -13.6$ eV

$\therefore \Delta E = E_2 - E_1 = 54.4 - 13.6 = 40.8$ eV

$\therefore K_1 + K_2 = K - \Delta E$

$K_1 + K_2 = 60 - 40.8$

$\Rightarrow K_1 + K_2 = 19.2$ …(iv)

Solving (iii) and (iv), we get

$K_1 = 3.36$ eV and $K_2 = 15.84$ eV

4. D

Sol. Let V is emf of the battery

$$i = \frac{V}{r + R_1 + R_2} \quad \text{…(i)}$$

$$V_{AB} = \frac{VR_1}{r + R_1 + R_2} \quad \text{…(ii)}$$

$$V_{A'B'} = 8i\left(\frac{R_1R}{R_1 + R}\right) = \frac{V_{AB}}{2} \quad \text{…(iii)}$$

From (i), (ii) and (iii)

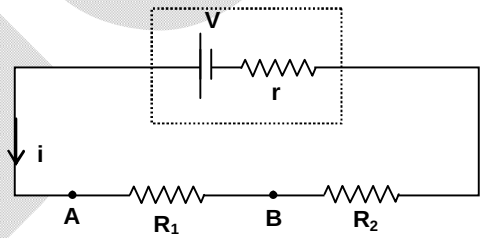
$R_1 = 150\ \Omega$

From (ii) circuit

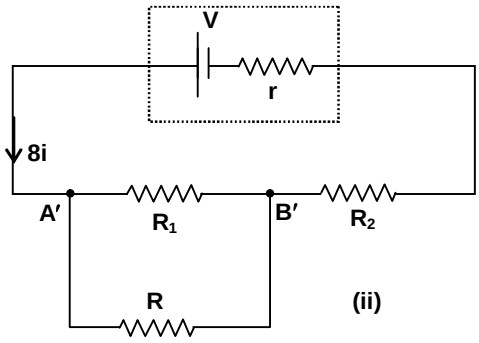
$$8i = \frac{V}{\left(r + R_2 + \frac{R_1R}{R_1 + R}\right)} \quad \text{…(iv)}$$

From (i) and (iv)

$R_2 = \frac{5}{7}\Omega$



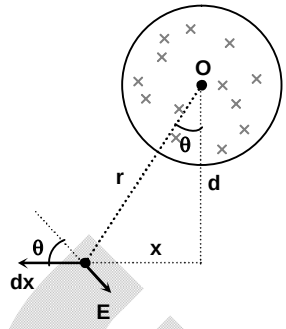
(i)



(ii)

5. B

Sol. $\int \vec{E}\cdot d\vec{\ell} = A\cdot\frac{dB}{dt}$



$E\cdot 2\pi\sqrt{x^2+d^2} = \pi R^2 K$, where $\left(\frac{dB}{dt}=K\right)$

$E = \frac{R^2K}{2\sqrt{x^2+d^2}}$

$W = \int_0^8 q\vec{E}\cdot d\vec{x}$

$= \int_0^8 qE\cos\theta\cdot dx = \int_0^8 \frac{qR^2k}{2\sqrt{x^2+d^2}}\frac{d}{\sqrt{d^2+x^2}}dx = \frac{qR^2K\pi}{8}$

Alternate solution

By applying Faraday's Law for an imaginary triangular loop having vertex O, P and point at infinity, $q\int \vec{E}\cdot d\vec{\ell}$ can be found out.

6. C

Sol. $F_t v = c = 10$

$m\frac{dv}{dt}v = c$

$\Rightarrow \int_0^t dt = \frac{m}{c}\int_{v_1}^{v_2} vdv$

$t = \frac{m}{2c}(v_2^2 - v_1^2) = \frac{m}{2c}\left(\frac{GM}{R} - \frac{GM}{3R}\right) = \frac{GMm}{2c}\left(\frac{1}{R} - \frac{1}{3R}\right) = \frac{GMm}{30R}$

7. A

Sol. from centre of mass frame

$N = \frac{4m(v_0 - v_{cm})^2}{r_{cm}}$ …(i)

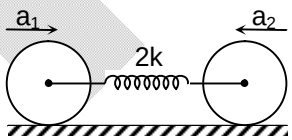
$v_{cm} = \frac{4mv_0 + 6m\times 0}{6m+4m} = \frac{2}{5}v_0$ …(ii)

$r_{cm} = \frac{4m\times 0 + 6m\cdot R}{6m+4m} = \frac{3}{5}R$ …(iii)

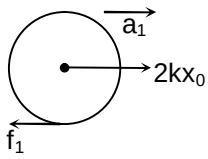
On solving $N = \frac{12}{5}\frac{mv_0^2}{R}$

8. A

Sol.



For hollow sphere



$2kx_0 - f_1 = ma_1$ …(i)

$2kx_0R = \frac{5}{3}mR^2\alpha$

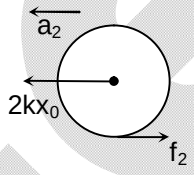
$R\alpha = a_1 = \frac{6}{5}\frac{kx_0}{m}$ …(ii)

From (i) and (ii)

$f_1 = 2kx_0 - \frac{6}{5}kx_0$

$f_1 = \frac{4}{5}kx_0$

For solid sphere



$2kx_0 - f_2 = ma_2$ …(iii)

$2kx_0R = \frac{7}{5}mR^2\alpha$

$R\alpha = a_2 = \frac{10}{7}\frac{kx_0}{m}$ …(iv)

From (iii) and (iv)

$f_2 = 2kx_0 - \frac{10}{7}kx_0$

$f_2 = \frac{4}{7}kx_0$

$\vec{a}_{cm} = \frac{m\vec{a}_1 + m\vec{a}_2}{m+m} = \left(\frac{6}{5} - \frac{10}{7}\right)\hat{i}\frac{kx_0}{2m} = \frac{-4}{35}\frac{kx_0}{m}\hat{i}$

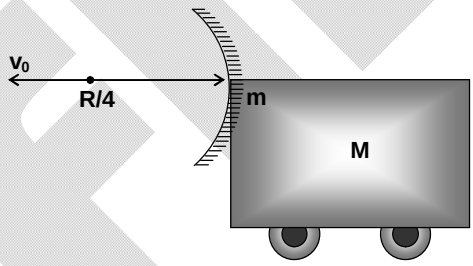
9. A

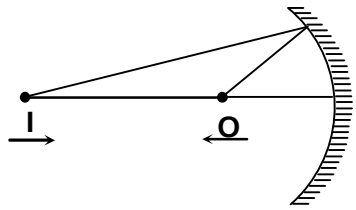
Sol. Let mass of the tank is M, mass of the bomb is m and v_0 is the velocity of the bomb w.r.t of the time.

$v_{IP} = -m_t^2 v_{OP} = -\left(\frac{f}{f-u}\right)^2 v_{OP}$

$v_{IP} = +4v_0$

$v_{PE} = \left(\frac{mv_0}{m+M}\right)$





$\vec{v}_{IP} = \vec{v}_{IP} + \vec{v}_{PG}$

$= 4v_0\hat{i} + \left(\frac{mv_0}{M+m}\right)\hat{i} = \frac{(4M+5m)}{(m+M)}v_0\hat{i}$

$\vec{v}_{IP} = \vec{v}_{IG} - \vec{v}_{OG}$

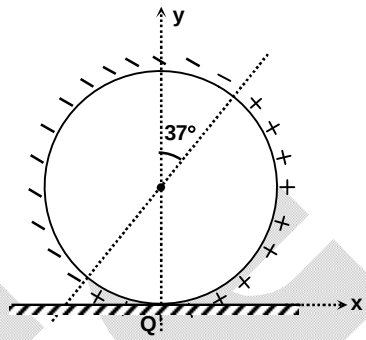
$= \frac{(4M+5m)}{(m+M)}v_0\hat{i} + \left(\frac{mv_0}{m+M}\right)\hat{i} = \frac{(5M+5m)}{(m+M)}v_0\hat{i} = 5v_0\hat{i}$

$$\vec{v}_{MO} = \vec{v}_{MG} - \vec{v}_{OG}$$
$$= \frac{mv_0}{m+M}\hat{i} + \frac{Mv_0\hat{i}}{m+M} = v_0\hat{i}$$

10. C

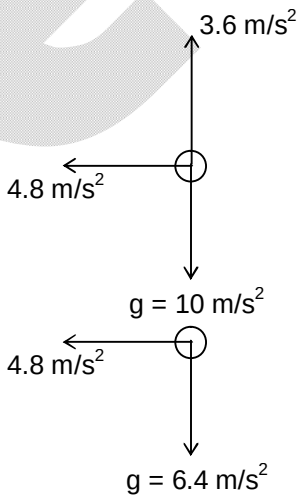
Sol. Dipole moment = $= \left(\rho\frac{2}{3}\pi R^3\right)\left(\frac{3R}{4}\right)$

$$\tau_Q = PE\sin 53^\circ$$
$$= \left(\rho\frac{2}{3}\pi R^3\right)\left(\frac{3R}{4}\right)E_0\frac{4}{5}$$
$$= \rho\frac{2}{3}\pi R^3\left(\frac{3R}{4}\right)\left(\frac{4}{5}\right)E_0 = \left(\frac{7}{5}mR^2\alpha\right)$$
$$\alpha = \frac{2}{7}\frac{\pi\rho R^2 E_0}{m}$$



11. B

Sol. effective g = $\sqrt{(4.8)^2 + (6.4)^2} = 8$ m/s^2



$$T = v\rho g_{eff} - v\frac{\rho}{2}g_{eff} = \frac{v\rho g_{eff}}{2} = \frac{2\times10^{-3}\times10^3\times8}{2} = 8$$

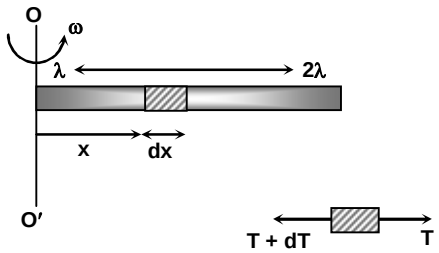
SECTION – B

12. 64

Sol. $$-\int_0^T dT = \int_\ell^x \left[\lambda\left(1+\frac{x}{\ell}\right)dx\right]x\omega^2$$
$$T = \lambda\omega^2\left(\frac{5\ell^2}{6} - \frac{x^2}{2} - \frac{x^3}{3\ell}\right)$$

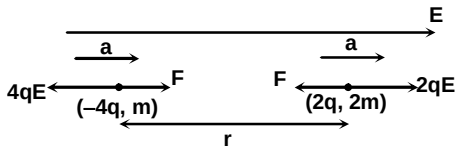
Let increment in length in dx is dδ.

$$\int_0^\delta d\delta = -\int_0^\ell \frac{\lambda\omega^2}{AY}\left(\frac{5\ell^2}{6} - \frac{x^2}{2} - \frac{x^3}{3\ell}\right)dx$$
$$\delta = \frac{7\lambda\omega^2\ell^3}{12AY} = 64\times10^{-5}\text{ m}$$



13. 8

Sol. On releasing the particles, if their acceleration is same, they will stay at a constant distance



Acceleration of charged particle

$$a = \frac{F - 4qE}{m} \quad \ldots(i)$$

$$a = \frac{2qE - F}{2m} \quad \ldots(ii)$$

From equation (i) and equation (ii)

$$F = \frac{10}{3}qE$$

When they stay at a constant distance from each other

$$\frac{8kq^2}{r^2} = \frac{10qE}{3} \Rightarrow r = \sqrt{\frac{12kq}{5E}}$$

$$= \frac{-k8q^2}{r} = -8 \times \sqrt{\frac{5kEq^3}{12}} = -8\ J$$

14. 5625

Sol. $X_C = \frac{V_C}{I} = \frac{4000}{2} = 2000\Omega$

Current lags behind voltage

So, box has an inductor

Power factor = 0.6 = $\frac{R}{Z} = \frac{1200}{\sqrt{(1200)^2 + (X_L - 2000)^2}}$

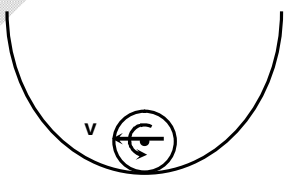
$X_L = 3600\ \Omega$

$\omega L = 3600$

$$L = \frac{3600}{2 \times 3.2 \times 100} = 5.625\ H$$

15. 140

Sol. Let parabolic equation is



$x^2 = ky$

$140 = k.7 \Rightarrow k = 20$

$$y = \frac{x^2}{20}$$

$$R = \frac{\left[1 + \left(\frac{dy}{dx}\right)^2\right]^{3/2}}{\left(\frac{d^2y}{dx^2}\right)} = 10m$$

By the conservation of energy

$$mgh = \frac{1}{2}mv^2 + \frac{1}{2}\left(\frac{2}{3}mR^2\omega^2\right) = \frac{5}{6}mv^2$$

$$mgh = \frac{5}{6}mv^2 \Rightarrow v = \sqrt{\frac{6}{5}gh} = \sqrt{60}\ m/s$$

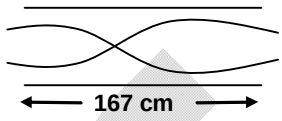
So $a_{centripetal} = \frac{v^2}{R} = 7.5\ m/s^2$

and $N - mg = \frac{mv^2}{R}$

$N = mg + \frac{mv^2}{R} = 80 + 60 = 140N$

16. 450

Sol. $\frac{\lambda}{2} = 167 + 0.6D$ …(i)



$v = f\lambda$

$\lambda = \frac{340}{100} = 340$ cm …(ii)

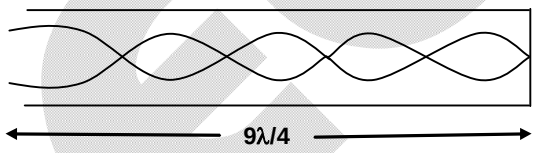
From (i) and (ii)

$\frac{340}{2} = 167 + 0.6D$

D = 5 cm

The frequency in the fifth harmonic

$\frac{5\lambda}{4} = 167 + 0.3D = 167 + 3 = 170$



$f = \frac{v}{\lambda} = 450$ Hz

17. 180

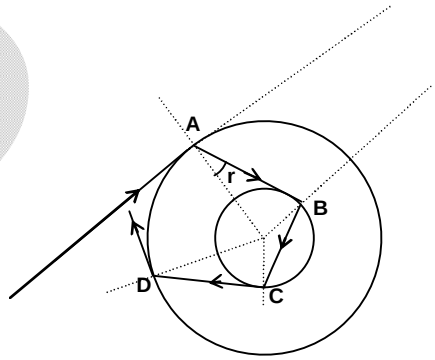
Sol. $1 \sin 90° = \frac{5}{3} \sin r$

$r = 37°$

For inner surface

Ray AB also incident at angle of grazing incidence

$\therefore$ Total angel of deviation = 53°+37°+37°+53° = 180°



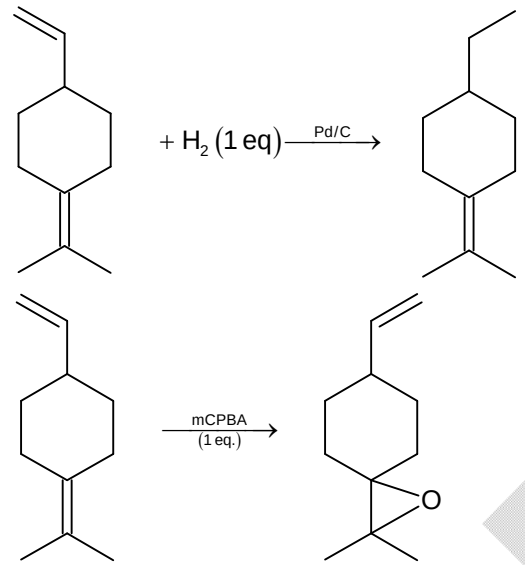
Chemistry

PART – II

SECTION – A

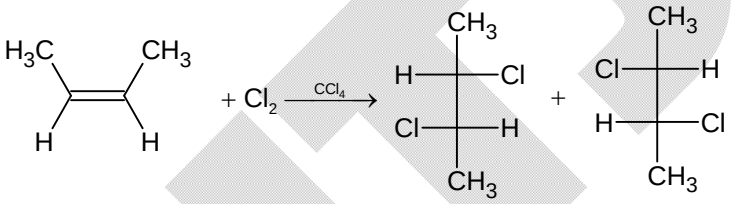
18. B, C

Sol.



19. A, B, C

Sol.



20. A, B, C, D

Sol. for spontaneous process at constant temperature, pressure, $\Delta G < 0$ and for adiabatic irreversible process.

$[\Delta S]_{sys} > 0,\ [\Delta S]_{total} > 0$ and $[\Delta S]_{surr} = \frac{q}{T} = 0$

21. A

Sol. In B, C and D co-ordination isomerism is possible.

22. A

Sol. E_{cell} is intensive property and mass, internal energy, enthalpy is extensive property.

23. A

Sol.

1.	RCR	2.	SCS
3.	SCR	4.	RTR
5.	STS	6.	STR

Total number of stereo isomers = 6

Number of enantiomeric pair = 2

24. A

Sol. $PCl_5 + H_2O \longrightarrow \underset{\text{(triprotic oxoacid)}}{H_3PO_4} + HCl$

25. B

Sol. Bond order $= \frac{1}{2}[N_b - N_a]$

In O_2, S and P mixing is not operative.

26. A

Sol. In $Si_2O_7^{6-}$ number of shared oxygen.

Atom per tetrahedral unit = 1

In SiO_2 , number of shared oxygen atom per tetrahedral unit = 4

27. A

Sol. P, Q, R and D are monodented ligand]

Number of stereo isomers = n

$(n)_{Ma_3b_3} = 2, \quad (n)_{Ma_2b_2c_2} = 6$

$(n)_{Ma_3bcd} = 5, \quad (n)_{Ma_3b_2c} = 3$

28. A

Sol. $[Fe(CN)_6]^{4-} \rightarrow d^2sp^3, \text{diamagnetic}$

$[Fe(CN)_6]^{3-} \rightarrow d^2sp^3, \text{paramagnetic}$

$[Cu(NH_3)_4]^{2+} \rightarrow dsp^2, \text{paramagnetic}$

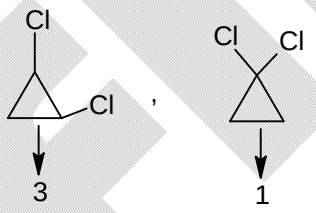
$[Ni(Cl)_4]^{2-} \rightarrow sp^3, \text{tetrahedral}$

SECTION – B

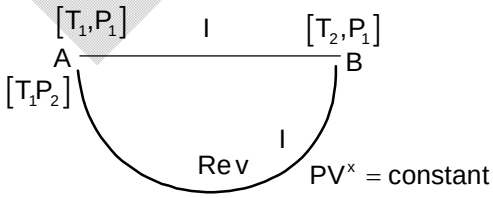
29. 2

Sol. From approach 2 sigma complex is more stable due to more stable resonating structure.

30. 4

Sol. 

31. 12

Sol. 

$$\left[\Delta S\right]_I = nC_{P,m}\ln\frac{T_2}{T_1} \quad \ldots (1)$$

$$\left[\Delta S\right]_{II} = nC_{P,m}\ln\frac{T_1}{T_2} + nR\ln\frac{P_1}{P_2} \quad \ldots (2)$$

By using $PV^x = \text{constant}$

$$\ln\frac{P_1}{P_2} = \frac{x}{1-x}\ln\frac{T_1}{T_2}$$

$$\therefore \left[\Delta S\right]_{II} = nC_{P,M}\ln\frac{T_1}{T_2} + (nR)\left(\frac{x}{1-x}\right)\ln\frac{T_1}{T_2}$$

$$\therefore \left[\Delta S\right]_I = \left[\Delta S\right]_{II}$$

$$nC_{P,M}\ln\frac{T_2}{T_1} = nC_{P,M}\ln\frac{T_1}{T_2} + \left[\frac{nR}{1-x}\right]x\ln\frac{T_1}{T_2}$$

$$\therefore x = \frac{6}{5}$$

$$\therefore P = \frac{6}{5}$$

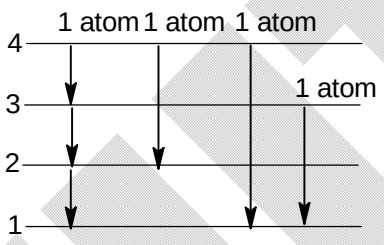
$$\therefore 10P = \frac{6}{5}\times 10 = 12$$

32. 4

Sol. NiS, CoS, PbS and Cus are black colour sulphide.

33. 4

Sol. Number of spectral lines $= \frac{n(n-1)}{2} = \frac{4\times 3}{2} = 6$



34. 160

Sol. Molar mass of ethylene glycol = 62 g/mol

$$\Delta T_f = K_f \times m$$

$$5 = 1.86\times\frac{\frac{40}{62}}{W}\times 1000$$

$$5 = 1.86\times\frac{40}{62}\times\frac{1000}{W}$$

$$W_{H_2O} = 1.86\times\frac{40}{62\times 5}\times 1000$$

$$W_{H_2O} = 240 \text{ g}$$

$\therefore$ The separated out = 400 – 240 = 160 g

Mathematics

PART – III

SECTION – A

35. B, C

Sol. $h_2(x) = \min\limits_{0\le t\le x}\left(t^2 + (x-t)^2\right) = \min\limits_{0\le t\le x}\left(2t^2 - 2tx + x^2\right) = \dfrac{x^2}{2}$

Similarly, $h_3(x) = \min\limits_{0\le t\le x}\left(\dfrac{t^2}{2} + (x-t)^2\right) = \dfrac{x^2}{3}$.....; $h_n(x) = \dfrac{x^2}{n}$

36. B, C, D

Sol. $\dfrac{\tan^4 x + \tan^4 y + \cot^2 x\cot^2 y + \cot^2 x\cot^2 y}{4} \ge 1$ (Using AM $\ge$ GM)

$\Rightarrow$ LHS ≥ 4 and RHS ≤ 4

$\therefore$ Solution possible only if LHS = RHS = 4

$\Rightarrow x + y = (2n+1)\dfrac{\pi}{2}$ and $x = \left\{\dfrac{\pi}{4}, \dfrac{3\pi}{4}, \dfrac{5\pi}{4}, \dfrac{7\pi}{4}\right\}$ and $y = \left\{\dfrac{\pi}{4}, \dfrac{3\pi}{4}, \dfrac{5\pi}{4}, \dfrac{7\pi}{4}\right\}$

37. B, C

Sol. Let $|5\sin x - 12\cos x| = t$

$\Rightarrow |t|^2 - (k^2 + k + 5)|t| + k^3 + 3k^2 + 2k + 6 = 0$

$\Rightarrow |t| = \dfrac{(k^2+k+5) \pm \sqrt{(k^2+k+5)^2 - 4(k^3+3k^2+2k+6)}}{2}$

$\Rightarrow |t| = k^2 + 2,\ k + 3$ for real solution

$0 \le k^2 + 2 \le 13$ or $0 \le k + 3 \le 13$

$\Rightarrow -\sqrt{11} \le k \le \sqrt{11}$ or $-3 \le k \le 10$

38. B

Sol. $f(1) = f(5)$

$\log_e\left(\dfrac{1+k}{8}\right) = \log_e\left(\dfrac{25+k}{40}\right) \Rightarrow k = 5$

$f'(x) = \dfrac{2x}{x^2+5} - \dfrac{1}{x} = 0 \Rightarrow c = \sqrt{5}$

$f''(x) = \dfrac{10 - 2x^2}{(x^2+5)^2} + \dfrac{1}{x^2} \Rightarrow f''(c) = \dfrac{1}{5}$

39. D

Sol. Put $y = 1$; $f(x) - f(x+1) = -f(1) - x$

Put $x = 1, 2, 3, ..., n-1$

$\Rightarrow f(1) - f(n) = -(n-1)f(1) - \dfrac{n(n-1)}{2}$

$\Rightarrow f(n) = \dfrac{n(n-1)}{2}$

40. B

Sol. If all powers are added, we get $1 + 2 + 3 + ... + 10 = 55$ i.e. 6 more than 49

Case-I: Excluding x^6 term $(x-1)(x^2-2)(x^3-2^2)$ ..... $(x^{10} - 2^9)(x^6 - 2^5)$

$\Rightarrow$ Coefficient of $x^{49} = -32$

Case-II: Excluding x^1 and x^5 terms $(x^2-2)(x^3-2^2)$ ..... $(x^{10} - 2^9)(x-1)(x^5 - 16)$

$\Rightarrow$ Coefficient of $x^{49} = 1 \times 16 = 16$

Case-III: Excluding x^2 and x^4 terms $(x-1)(x^3-2^2)(x^5-2^4)(x^6-2^5)$ ..... $(x^{10}-2^9)(x^2-2)(x^4-2^3)$

$\Rightarrow$ Coefficient of $x^{49} = -2 \times -8 = 16$

Case-IV: Excluding x^1, x^2 and x^3 terms $(x^9-2^3)(x^5-2^4)$ ..... $(x^{10}-2^9)(x-1)(x^2-2)(x^3-4^2)$

$\Rightarrow$ Coefficient of $x^{49} = -1 \times -2 \times -2^2 = -8$

$\therefore$ Coefficient of x^{49} in given expansion $= -32 + 16 + 16 - 8 = -8$

41. A

Sol. Let the number be $x_1\,x_2\,x_3\,x_4\,x_5$ where $0 \le x_1 \le 9$, $0 \le x_2 \le 9$, $0 \le x_3 \le 9$, $0 \le x_4 \le 9$, $0 \le x_5 \le 9$ and $x_1 + x_2 + x_3 + x_4 + x_5 = 15$

Number of solutions = Coefficient of x^{15} in $(1 + x + x^2 + ... + x^9)^5$

= Coefficient of x^{15} in $(1-x^{10})^5(1-x)^{-5} = {}^{19}C_4 - 5 \times {}^{9}C_4 = 3246$

42. B

Sol. (P) $f(x) = \int_0^x \frac{1}{1+x-t}dt + \int_x^1 \frac{1}{1+t-x}dt = -\ln|1+x-t|_0^x + \ln|1+t-x|_x^1 = \ln(1+x) + \ln(2-x)$

(Q) $\int_0^1 \left(f(x) - f^2(x)\right)^2 dx = 0 \Rightarrow f(x) = f^2(x) \Rightarrow f(x) = 0,\ f(x) = 1$

(R) $\int_0^2 \left(f^2(x) - 3x\right)^2 = 0 \Rightarrow f^2(x) = 3x$

(S) $\left[\int_0^{1/\sqrt{3}} 3dx + \int_{1/\sqrt{3}}^{1} 2dx + \int_1^{\sqrt{3}} 1dx\right] = \left[1 + \frac{4}{\sqrt{3}}\right] = 3$

43. C

Sol. (P) $2(\sin^4\theta_1 + \cos^4\theta_1) + 4 + 2(\text{cosec}^4\theta_1 + \sec^4\theta_1) + 4 - 24 = \sin\theta_2$

$= 2\left(1 - \frac{1}{2}\sin^2\theta_1\right)\left(1 + \frac{16}{\sin^4\theta_1}\right) = 16 + y$

LHS is ≥ 17 and RHS ≤ 17

Solution possible only LHS = RHS = 17 $\Rightarrow \sin^2 2x = 1$ and $\sin y = 1$

$\Rightarrow x = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4} \Rightarrow y = \frac{\pi}{2}, \frac{5\pi}{2}$

(Q) LHS ≤ 2 and RHS ≥ 2 but both can't be 2 simultaneously

(R) Let $(1 - \sin t) = u$

$\Rightarrow \frac{du}{dt} = -\cos t$

$\therefore\ k = \sum_{n=1}^{8} 2\int_0^1 u^n(1-u)d = 2\sum_{n=1}^{8}\left(\frac{1}{n+1} - \frac{1}{n+2}\right) = 2\left(\frac{1}{2} - \frac{1}{10}\right) = \frac{4}{5}$

(S) $6\left(\frac{1-\tan^2\frac{\theta}{2}}{1+\tan^2\frac{\theta}{2}}\right) + 8\left(\frac{2\tan\frac{\theta}{2}}{1+\tan^2\frac{\theta}{2}}\right) = 9 \Rightarrow 15\tan^2\frac{\theta}{2} - 16\tan\frac{\theta}{2} + 3 = 0$

$\Rightarrow \tan\frac{\alpha}{2} + \tan\frac{\beta}{2} = \frac{16}{15}$ and $\tan\frac{\alpha}{2}\tan\frac{\beta}{2} = \frac{3}{15}$

44. B

Sol. (P) D.C. of hyperbola $15(x-3)^2 - 16(y-2)^2 = 240$ is $(x-3)^2 + (y-2)^2 = 1$ $\therefore$ h = 3

(Q) Required points are point of intersection of $xy = c^2$ and $x^2 + y^2 = a^2 + b^2$

$\Rightarrow x^2 + \frac{c^4}{x^2} = a^2 + b^2 \Rightarrow x^4 - (a^2 + b^2)x^2 + c^4 = 0$

$\because$ D < 0 $\therefore$ No solution hence no point intersection

(R) $C_1 : x^2 + y^2 - 2x - 15 = 0$

$C_2 : (x-7)^2 + (y-8)^2 = a^2 + 20$

Distance between centres must be equal to the sum of radii

$\therefore 10 = 4 + \sqrt{a^2 + 20} \Rightarrow a = 4$

(S) $(\lambda, 2)$ must be on directrix of $(y-3)^2 = 16(x-5)$ $\therefore \lambda = 1$

45. A

Sol. (P) Mean $\frac{100 + 5k}{18 + k}$ $\Rightarrow$ if k = 7 mean = 5.4

(Q) $\frac{100 + 5k}{18 + k} = 5.5 \Rightarrow k = 2$

(R) $\bar{x} = \frac{110}{20} = 5.5$

$\therefore$ Mean deviation about mean = $\frac{(4\times 3.5) + (8\times 1.5) + (3\times 3.5) + (2\times 0.5) + (3\times 2.5)}{20}$ = 2.25

(S) $\frac{100 + 5k}{18 + k} = 5.25 \Rightarrow k = 22$ $\therefore$ Mode = 5

SECTION – B

46. 85

Sol. Let 6 numbers be $x_1, x_2, x_3, x_4, x_5, x_6$ such $|x_i - x_j| \geq 8$

Let d_1 be the numbers before x_1
Let d_2 be the numbers between x_1 and x_2
Let d_3 be the numbers between x_2 and x_3
Let d_4 be the numbers between x_3 and x_4
Let d_5 be the numbers between x_4 and x_5
Let d_6 be the numbers between x_5 and x_6
Let d_7 be the numbers after x_6
$d_1 + d_2 + d_3 + d_4 + d_5 + d_6 = 114$
$d_1 \geq 0, d_2 \geq 7, d_3 \geq 7, d_4 \geq 7, d_5 \geq 7, d_6 \geq 7, d_7 \geq 0$
Let $d_i = t_i + 7, t_i \geq 0, i \in \{2, 3, 4, 5, 6\}$
$d_1 + t_2 + t_3 + t_4 + t_5 + d_6 + d_7 = 79$
Number of non-negative integral solution = ${}^{79+7-1}C_{7-1} = {}^{85}C_6$

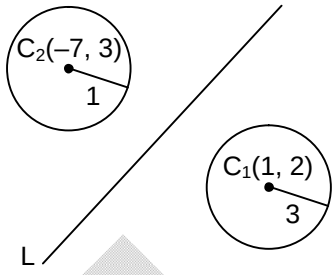
47. 16

Sol. Let $x = \frac{4t}{3+t} \Rightarrow \frac{dx}{dt} = \frac{12}{(3+t)^2}$

$(\text{In } I_1) \Rightarrow I_1 = \int_0^1 \frac{(4t)^{\frac{5}{2}}(3-3t)^{\frac{7}{2}}}{(3+t)^6} \times \frac{12}{(3+t)^2} dt = 4^{\frac{5}{2}} 3^{\frac{7}{2}} \times 12 I_2 = 2^7 3^{\frac{9}{2}}$

48. 7

Sol. The line doesn't touch or intersect either of the circles if perpendicular distance from (1 on L is greater than r_1 and perpendicular distance from C_2 on L is greater than r_2



$\Rightarrow \frac{|3-8+k|}{5} > 3 \Rightarrow k \in (-\infty, -10) \cup (20, \infty)$

and $\frac{|-2-12+k|}{5} > 1 \Rightarrow k \in (-\infty, 28) \cup (38, \infty)$

If the line L has to lie in between the circles, then C_1 and C_2 must be on the opposite side of L

$(k - 5)(k - 33) < 0 \Rightarrow k(5, 33)$

Taking intersection of all 3 intervals, we get $k \in (20, 28)$

49. 2023

Sol. $\int_0^x f(t)dt + \int_0^x (x-t)f(t)dt = e^{-x} - 1 \Rightarrow x\int_0^x f(t)dt + \int_0^x f(t)(1-t)dt = e^{-x} - 1$

Differentiating both sides wrt x

$\Rightarrow xf(x) + \int_0^x f(t)dt + f(x)(1-x) = e^{-x}$

Differentiating both sides wrt x

$\Rightarrow f(x) + f'(x) = e^{-2} \Rightarrow f(x)e^x = x - 1$

50. 8

Sol. $\left(\sum_{i=1}^{12} p_i\right)^2 = \sum_{i=1}^{12} p_i^2 + 2\sum p_k p_j$

Variance = $\frac{\sum_{i=1}^{12} p_i^2}{12} - (\text{mean})^2 = \frac{528}{12} - 36 = 8$

51. 78

Sol. ${}^6C_0 \times {}^{60}C_6 + {}^6C_1 \times {}^{60}C_7 + ... + {}^6C_6 \times {}^{60}C_{12}$

Coefficient of x^{54} in $= (1 + x)^6(x + 1)^{60}$ = Coefficient of x^{54} in $(1 + x)^{66}$

$= {}^{66}C_{54} = {}^{66}C_{12}$